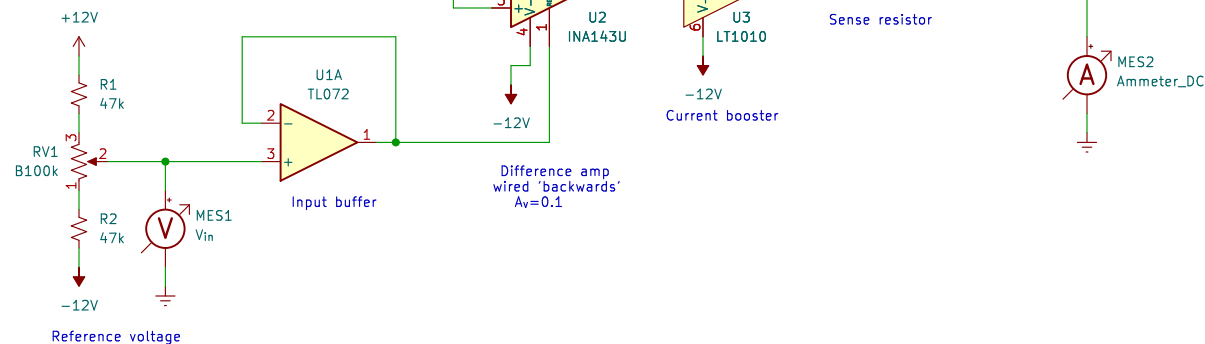


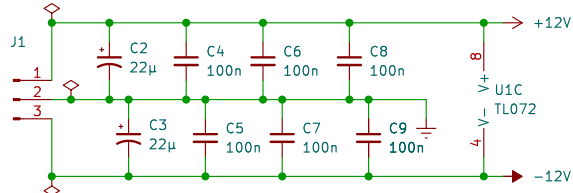


The change from the second version is the addition of U3, a high-current buffer amp. The buffer is inside the feedback loop, so any gain error or offset in the buffer should be compensated for.

Sense resistor has been decreased to 10 ohms, for a current range of $-60 \dots 60$ mA.

All resistors are 1/4 W 1% metal film except as noted.
 * – See description for the values used in the different tests.
 C2 and C3 are 25V aluminum electrolytic; all others are 50V X5R or Y7V ceramic.

Power and decoupling



Place one pair of 100 nF capacitors near the power pins of each IC



Demonstration repeated with addition of feedback buffer.
 Circuit now performs well with microamp-level currents and a 10k sense resistor.

Op-Amp Basics
 Episode 17 – Transconductance Amplifiers 4
Kludges from Kevin's Cave
 Sheet: /
 File: Transconductance4-Widlar-3.kicad_sch

Title: Widlar Current Pump

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